LSSO: Learnable Low-Rank Sylvester Solves for Efficient Bidirectional Encoders

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Abstract

We introduce LSSO, a token mixer for bidirectional encoders built around a learned low-rank linear solve. Rather than routing values through an explicit pairwise attention matrix, LSSO constructs an input-dependent relation field and solves

$$(\mu I + \gamma U(X)U(X)^\top)Y = C(X).$$

This is the left-sided case of a Sylvester equation. The positive shift makes the system well posed, while the Woodbury identity reduces the computation to an $r \times r$ solve. For fixed relation features, the operator has a bounded spectral response and minimizes a strongly convex correction energy. Across retrieval, MS MARCO-to-BEIR transfer, and vision classification, LSSO matches or approaches attention-based encoders while using substantially fewer mixer MACs. Ablations, rank pruning, and sequence-scaling experiments further show when the global correction is useful and where the current formulation falls short.

1 Introduction

Self-attention [21] is the standard token mixer in modern bidirectional encoders. Given query, key, and value projections, it computes

$$Y_{\text{attn}} = \text{softmax}(QK^\top/\sqrt{d})V,$$

and lets every token route information from every other token. That flexibility comes with quadratic score and value-routing costs. Most efficient attention methods reduce the cost by approximating or sparsifying the routing matrix. LSSO asks a different question: can a global token interaction be learned without constructing such a matrix at all?

For an input sequence $X \in \mathbb{R}^{B\times N\times D}$, LSSO learns a low-rank relation feature $U(X)$ and a content state $C(X)$. For each head it solves

$$(\mu I_N + \gamma UU^\top)Y = C, \qquad \mu > 0, \quad \gamma \geq 0,$$

and projects the result back to the model dimension. Although $UU^\top$ is positive semidefinite, its entries $u_i^\top u_j$ may be positive or negative. The system can therefore encode signed token relations while remaining invertible by construction.

The guiding interpretation is:

$$\begin{aligned}\text{LSSO} &= \text{isotropic content pass-through}\\ &\quad - \text{low-rank global correction.}\end{aligned}$$

Attention learns a row-normalized routing rule; LSSO learns the relation features of a positive-definite inverse filter. The two mixers therefore have different inductive biases even when their input and output shapes match.

The main contributions are:

1. We formulate bidirectional token mixing as a learned, positive-shifted low-rank solve whose implementation only solves an $r \times r$ system.

2. We characterize the fixed-U operator through its spectrum, conditioning, norm bound, and equivalent convex energy.

3. We evaluate the mixer on retrieval, zero-shot transfer, vision, rank pruning, diagnostics, and long-sequence scaling, including settings where LSSO does not outperform attention.

2 Related Work

Efficient attention. Linformer [23], Performer [5], Nyströmformer [25], Longformer [3], and BigBird [27] reduce the cost of self-attention through low-rank projections, kernel approximations, landmark approximations, or sparse attention patterns. Despite these differences, they retain attention as the underlying interaction rule. LSSO instead replaces that rule with an input-dependent inverse filter.

Token mixers beyond attention. MLP-Mixer [19], gMLP [14], FNet [13], Hyena [17], and state-space models such as Mamba [11] demonstrate that useful token mixing does not require dense attention. LSSO explores a complementary construction: an implicit low-rank solve designed specifically for non-causal encoders.

Implicit layers and learned solves. Deep equilibrium models [1], implicit layers, graph smoothing, and regularized least-squares methods place linear or nonlinear solves inside neural networks. LSSO is narrower than a general equilibrium layer: it applies one input-conditioned linear solve with a deliberately constrained positive-definite matrix.

3 Method

3.1 Encoder Block

LSSO is used in the token-mixer slot of an encoder block. In a pre-normalized Transformer-style block this is simply

$$X' = X + \mathrm{LSSO}(\mathrm{LN}(X)), \qquad X_{\text{out}} = X' + \mathrm{MLP}(\mathrm{LN}(X')).$$

The replacement is local: the input and output shapes match MHA, so the same residual path, normalization, positional encoding, feed-forward block, and task head can be kept. In the BERT-style retrieval experiments we preserve the post-LN BERT block used by the baseline; in the ViT-style vision experiments we use the pre-LN form above. Thus the comparison changes the mixer, not the surrounding block convention.

3.2 LSSO Definition

For each head, let

$$U = XW_u \in \mathbb{R}^{B\times N\times r}, \qquad C = XW_c \in \mathbb{R}^{B\times N\times d_h}.$$

There is one pair of scalar solve parameters per head. We RMS-normalize U along its rank dimension by default,

$$U \leftarrow U/\sqrt{\mathrm{mean}_r(U^2)+\epsilon},$$

and solve

$$(\mu I_N + \gamma UU^\top)Y = C, \tag{1}$$

where

$$\mu = \mathrm{softplus}(\theta_\mu) + \epsilon, \qquad \gamma = \gamma_{\max}\sigma(\theta_\gamma).$$

The default experiments use $\gamma_{\max} = 0.3$ and $\theta_\gamma = -4$, which gives an initial γ/μ of roughly 7.8×10^{-3} when $\theta_\mu = 0$. More conservative settings such as $\gamma_{\max} = 0.1, \theta_\gamma = -6$ make the global correction nearly vanish at initialization; in preliminary runs they consistently underused the global term.

The head outputs are concatenated and projected:

$$\mathrm{LSSO}(X) = YW_o.$$

For padded sequences, invalid tokens are masked by zeroing the corresponding rows of both U and C before the solve and by zeroing the output at masked positions. This mirrors key-padding behavior in bidirectional attention while preserving the low-rank Woodbury structure over the valid tokens.

3.3 Woodbury Implementation

The $N\times N$ matrix in Eq. (1) never needs to be formed or inverted. Applying the Woodbury identity [24] gives

$$Y = \mu^{-1}C - \mu^{-1}U\left(I_r + \gamma\mu^{-1}U^\top U\right)^{-1}\left(\gamma\mu^{-1}U^\top C\right). \tag{2}$$

This expression remains valid at $\gamma = 0$ and requires only an $r\times r$ solve; Appendix A gives the full derivation. The released implementation fuses the U and C projections into one linear map for efficiency, but this is algebraically identical to using separate W_u and W_c. The small positive-definite system is solved by Cholesky in float32 under fp16/bf16 autocast, then cast back to the activation dtype. The current package also provides a custom autograd path whose backward pass reuses the same Woodbury solve on the adjoint, avoiding a dense $N\times N$ inverse in both directions. The core solve terms cost

$$O(Nr^2 + Nrd_h + r^2d_h + r^3)$$

per head, where r^2d_h accounts for applying the Cholesky factors to the d_h right-hand sides. Including projections, the mixer-only MAC estimate used in our tables is

$$ND(Hr+D) + ND^2 + HNr^2 + 2NrD + Dr^2 + Hr^3.$$

Here the final term conservatively counts each small factorization as r^3 MAC-equivalents. For MHA we count QKV projections, attention scores, attention-value product, and output projection:

$$4ND^2 + 2N^2D.$$

4 Theoretical Properties

We first study the linear map induced by a fixed U; the input-dependent operator is evaluated empirically in Section 5.

Proposition 1 (Well-posedness)**.** *Let $A = \mu I_N + \gamma UU^\top$ with $\mu > 0$ and $\gamma \geq 0$. Then A is symmetric positive definite. Therefore, for any C, the system $AY = C$ has a unique solution.*

Proof. For any nonzero z,

$$z^\top Az = \mu\|z\|_2^2 + \gamma\|U^\top z\|_2^2 \geq \mu\|z\|_2^2 > 0.$$

Proposition 2 (Spectral shrinkage)**.** *Let $UU^\top = Q\Lambda Q^\top$ be an eigendecomposition over the nonzero eigen-directions of $UU^\top$, with $\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_k)$ and $\lambda_i > 0$. Let $P_\perp = I - QQ^\top$. Then*

$$A^{-1} = Q\,\mathrm{diag}\left(\frac{1}{\mu+\gamma\lambda_i}\right)Q^\top + \mu^{-1}P_\perp.$$

Thus LSSO applies an inverse spectral filter on the span of U and an isotropic scale on its orthogonal complement. This gives a more precise view of the distinction from row-normalized attention.

Proposition 3 (Bounded fixed-U operator norm)**.** *For fixed U,*

$$\|Y\|_F \leq \mu^{-1}\|C\|_F.$$

For a multi-head layer, let Y_{cat} and C_{cat} denote the concatenated head outputs and inputs, and let $\mu_{\min} = \min_h \mu_h$. With the output projection applied on the right,

$$\|Y_{\text{cat}} W_o\|_F \leq \mu_{\min}^{-1}\|C_{\text{cat}}\|_F \|W_o\|_2.$$

Proof. Since $A \succeq \mu I$, $\lambda_{\min}(A) \geq \mu$ and $\|A^{-1}\|_2 \leq \mu^{-1}$. The Frobenius bound follows from $Y = A^{-1}C$. Summing the squared per-head bounds gives $\|Y_{\text{cat}}\|_F \leq \mu_{\min}^{-1}\|C_{\text{cat}}\|_F$, and $\|Y_{\text{cat}}W_o\|_F \leq \|Y_{\text{cat}}\|_F\|W_o\|_2$ completes the result.

Proposition 4 (Conditioning control)**.** *For $A = \mu I + \gamma UU^\top$,*

$$\kappa_2(A) \leq 1 + \frac{\gamma}{\mu}\|U\|_2^2.$$

Propositions 3 and 4 motivate the diagnostics used in our experiments. The ratio γ/μ directly scales the possible condition-number increase; effective rank measures how much of the rank subspace is used; and correction ratio measures how strongly the low-rank correction participates in the output. RMS normalization of U helps control the scale of U, although it is not by itself a dimension-free spectral-norm guarantee.

Theorem 1 (Energy minimization equivalence)**.** *For $\mu > 0$ and $\gamma \geq 0$, the LSSO output is the unique minimizer of*

$$\mathcal{E}(Y) = \frac{\mu}{2}\|Y\|_F^2 + \frac{\gamma}{2}\|U^\top Y\|_F^2 - \langle C, Y\rangle.$$

Proof. The Hessian of $\mathcal{E}$ is $\mu I + \gamma UU^\top$, which is positive definite by Proposition 1. Setting the gradient to zero gives $(\mu I + \gamma UU^\top)Y = C$.

Equivalently, up to an additive constant,

$$\mathcal{E}(Y) = \frac{\mu}{2}\left\|Y - \mu^{-1}C\right\|_F^2 + \frac{\gamma}{2}\|U^\top Y\|_F^2.$$

The content projection supplies the local state $\mu^{-1}C$, and the second term adjusts that state using sequence-wide information.

Proposition 5 (No-global degeneration)**.** *If $\gamma = 0$, then*

$$Y = \mu^{-1}C.$$

No token mixing occurs except for pointwise projections.

5 Experiments

5.1 Setup

All main retrieval and vision comparisons use randomly initialized encoders. The retrieval backbone follows a BERT-style bidirectional encoder layout [9]. We reuse standard tokenization but train every encoder weight from scratch. Models are optimized with AdamW [15]. LSSO uses $\gamma_{\max} = 0.3$ and $\theta_\gamma = -4$ unless stated otherwise. Because the feed-forward network is identical across models, tables report mixer-only MACs and isolate the cost of the component being replaced.

Retrieval protocol. The dual encoder has $D = 256$, depth 8, 8 heads, and dropout 0.1. Queries and documents are truncated to 64 and 512 tokens, respectively, and representations are obtained by masked mean pooling. We train with in-batch contrastive cross-entropy at temperature 0.05, using batch size 64, AdamW with learning rate 3×10^{-4} and weight decay 0.01, 5% linear warmup followed by cosine decay, and gradient clipping at 1.0. Each target-dataset run uses at most 50,000 training pairs for 20 epochs; the checkpoint with the highest validation R@10 is evaluated. All models see the same sampled pairs and use the same tokenizer, pooling rule, and three random seeds.

5.2 Retrieval Main Results

Table 1 reports 3-seed retrieval results on FIQA [16], NFCorpus [4], and SciFact [22], using their BEIR retrieval formulations [18]. The baseline is a BERT-style MHA encoder, and Nyströmformer uses the authors' attention module. LSSO-r16 and LSSO-r32 use the same model dimension, depth, number of heads, optimizer, and training schedule.

Table 1: Retrieval main results. Models are randomly initialized BERT-style encoders with $D = 256$, depth 8, heads 8, and document length 512. MACs are mixer-only document-side MACs.

Dataset	Model	Params (M)	Mixer MACs (G)	R@10	MRR@10
FIQA	MHA	14.33	2.147	0.2145 ± 0.0135	0.0987 ± 0.0032
FIQA	Nyströmformer	14.33	1.301	0.2428 ± 0.0039	0.1190 ± 0.0071
FIQA	LSSO-r16	13.54	0.714	0.2258 ± 0.0125	0.1126 ± 0.0061
FIQA	LSSO-r32	13.80	0.910	0.2387 ± 0.0183	0.1150 ± 0.0035
NFCorpus	MHA	14.33	2.147	0.5439 ± 0.0187	0.3809 ± 0.0145
NFCorpus	Nyströmformer	14.33	1.301	0.5841 ± 0.0125	0.4323 ± 0.0132
NFCorpus	LSSO-r16	13.54	0.714	0.5686 ± 0.0125	0.4121 ± 0.0036
NFCorpus	LSSO-r32	13.80	0.910	0.5841 ± 0.0036	0.4333 ± 0.0220
SciFact	MHA	14.33	2.147	0.6789 ± 0.0190	0.5994 ± 0.0138
SciFact	Nyströmformer	14.33	1.301	0.7267 ± 0.0133	0.6313 ± 0.0033
SciFact	LSSO-r16	13.54	0.714	0.7022 ± 0.0107	0.6214 ± 0.0119
SciFact	LSSO-r32	13.80	0.910	0.7089 ± 0.0117	0.6247 ± 0.0080

Relative to MHA, LSSO-r16 removes 66.8% of mixer MACs and LSSO-r32 removes 57.6%. Both improve the mean MHA result on every dataset in the table. Nyströmformer remains the strongest baseline on several metrics, although in our implementation its lower theoretical MAC count did not translate into higher measured throughput.

5.3 MS MARCO to BEIR Transfer

We next ask whether the retrieval result survives a change of domain. Each model is pretrained once on MS MARCO [2] and evaluated without target-task finetuning on five BEIR datasets [18] covering finance, biomedicine, scientific fact checking, argument retrieval, and COVID literature. Unlike the preceding experiments, this setting does not allow the encoder to adapt to each evaluation collection.

All four models use the same randomly initialized BERT-style architecture, MS MARCO training data, tokenizer, pooling, optimizer, three seeds, and document length 512; only the token mixer changes. For each seed, we first average a metric across the five target datasets and then report the mean and standard deviation of those macro averages. Since total parameter counts are similar (13.53–14.33M), Table 2 emphasizes mixer MACs. The released artifacts contain the corresponding per-dataset and per-seed values. Specifically, each model is trained for two epochs on 500,000 MS MARCO pairs with batch size 192, bf16 autocast, learning rate 3×10^{-4}, 5% warmup, and the same temperature and clipping settings as the target-dataset runs. Corpus embeddings are computed once per checkpoint and searched by inner product; no labels or optimizer steps from the five BEIR targets are used.

Table 2: Zero-shot BEIR transfer after MS MARCO pretraining. Retrieval metrics are five-dataset macro averages, reported as mean ± standard deviation over three seeds. MACs are mixer-only at document length 512. Best values are bold.

Model	Mixer MACs (G)	nDCG@10	Recall@100	MRR@10
MHA	2.147	0.22945±0.00208	0.36061±0.00365	**0.30225±0.00442**
Nyströmformer	1.270 (−40.9%)	0.20523±0.00495	0.33456±0.00837	0.27115±0.01046
LSSO-r16	**0.714** (−66.8%)	**0.22973±0.00232**	0.35889±0.00226	0.29382±0.01446
LSSO-r32	0.910 (−57.6%)	0.22940±0.00337	**0.36367±0.00830**	0.29858±0.00624

Both LSSO variants preserve MHA-level transfer at a much lower mixer cost. LSSO-r16 slightly exceeds the MHA macro nDCG@10 (0.22973 vs. 0.22945) while using 66.8% fewer mixer MACs. LSSO-r32 has the strongest macro Recall@100 (0.36367) while using 57.6% fewer mixer MACs. Nyströmformer is also cheaper than MHA, but trails both MHA and LSSO on all three macro metrics.

No single rank wins every metric. LSSO-r16 offers the best nDCG@10–cost trade-off, LSSO-r32 reaches the highest Recall@100, and MHA retains the best MRR@10. Taken together, the two LSSO settings lie on the same transfer frontier as MHA while requiring considerably less mixer computation.

5.4 Ablations and Diagnostics

Table 3 aligns the empirical ablations with the theory. The no-global variant sets $\gamma = 0$, reducing the operator to $\mu^{-1}C$ and removing token mixing as stated in Proposition 5. The fixed-scale variant freezes both θ_μ and θ_γ at initialization, while the no-normalization variant removes the row-wise RMS normalization of U. The r8 and r4 rows are independently trained full LSSO models at smaller rank, rather than pruned r32 checkpoints. Removing RMS normalization weakens the global correction, especially on SciFact, while the smaller-rank runs expose the capacity needed by the learned relation subspace.

Table 3: Retrieval ablations. Values are mean ± standard deviation over three seeds.

Dataset	Variant	R@10	MRR@10	γ/μ	Correction ratio
FIQA	no-global r32	0.2052 ± 0.0137	0.0911 ± 0.0019	0.0000	0.0000
FIQA	fixed μ/γ r32	0.2428 ± 0.0126	0.1136 ± 0.0028	0.0078	0.2130
FIQA	no U RMS norm r32	0.2160 ± 0.0101	0.1007 ± 0.0008	0.0085	0.2050
FIQA	full r8	0.2088 ± 0.0208	0.0923 ± 0.0139	0.0100	0.1059
FIQA	full r4	0.2042 ± 0.0045	0.0979 ± 0.0036	0.0109	0.0742
SciFact	no-global r32	0.6767 ± 0.0120	0.5811 ± 0.0058	0.0000	0.0000
SciFact	fixed μ/γ r32	0.7089 ± 0.0102	0.6088 ± 0.0066	0.0078	0.2631
SciFact	no U RMS norm r32	0.6933 ± 0.0133	0.5972 ± 0.0150	0.0078	0.0471
SciFact	full r8	0.7067 ± 0.0173	0.6118 ± 0.0064	0.0078	0.1149
SciFact	full r4	0.7022 ± 0.0069	0.6146 ± 0.0107	0.0078	0.0766

Figure 1 shows the same quantities layer by layer. The correction ratio remains nonzero on every task, so the trained models do not collapse to the local projection. Effective rank varies more substantially: SciFact uses a larger fraction of the rank-32 subspace than FIQA, NFCorpus, or CIFAR-100.

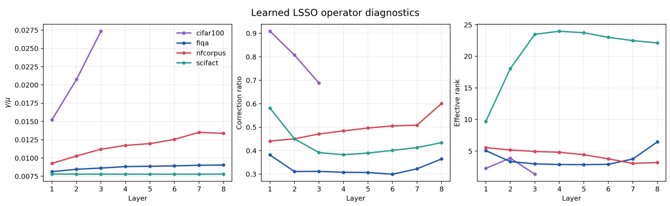

Figure 1: Layer-wise LSSO diagnostics from trained LSSO-r32 checkpoints.

5.5 Inference-Time Rank Pruning

The low effective ranks in Figure 1 suggest that a trained rank-32 model may not need every coordinate at inference. For each input and head, we rank coordinates by their mean squared activation in U and retain the top k before the Woodbury solve. Table 4 evaluates this procedure directly on the three-seed LSSO-r32 retrieval checkpoints, without finetuning after pruning.

Table 4: Inference-time pruning of trained LSSO-r32 models. "Save" is the theoretical mixer-MAC reduction of a compact rank-k implementation relative to rank 32; the current dynamic evaluator is used only to measure quality.

Keep rank	Save	FIQA R@10	NFCorpus R@10	SciFact R@10
32	0.0%	0.2387 ± 0.0183	0.5841 ± 0.0036	0.7089 ± 0.0117
16	21.6%	0.2377 ± 0.0107	0.5831 ± 0.0047	0.7056 ± 0.0135
8	31.5%	0.2361 ± 0.0149	0.5810 ± 0.0078	0.6922 ± 0.0139
4	36.3%	0.2269 ± 0.0157	0.5759 ± 0.0054	0.6856 ± 0.0139

Keeping 16 of 32 coordinates changes R@10 by less than 0.004 on every dataset while reducing the theoretical compact mixer cost by 21.6%. More aggressive pruning remains mild on FIQA and NFCorpus but degrades SciFact more clearly, consistent with its higher measured effective rank. These savings require exporting or executing the reduced-rank operator; merely masking coordinates inside the original rank-32 tensors does not reduce projection cost.

5.6 Sequence Scaling

We benchmark a single mixer layer at sequence lengths from 128 to 2048, using batch size 8, $D = 256$, 8 heads, bf16 autocast, and no optimizer update. Figure 2 reports forward latency, forward-plus-backward latency, incremental peak memory, and theoretical mixer MACs.

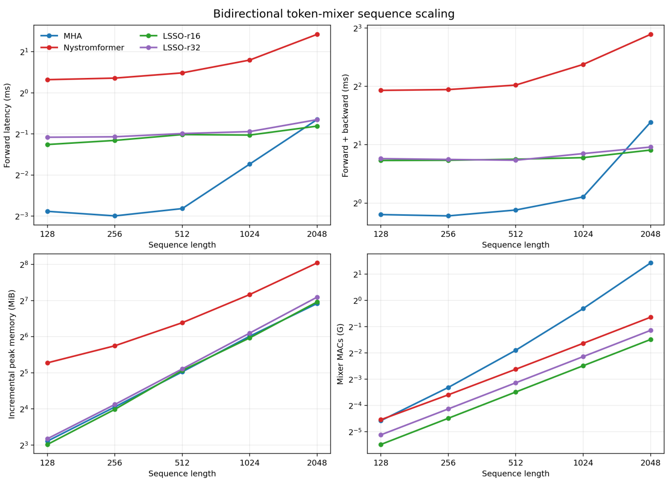

Figure 2: Sequence scaling benchmark on an NVIDIA RTX 5070 Ti. MACs are theoretical per-sample single-layer mixer MACs; latency and memory use batch 8.

At $N = 2048$, LSSO-r16 uses 0.357G mixer MACs compared with 2.684G for MHA, an 86.7% reduction, and reduces forward+backward latency from 2.61 ms to 1.88 ms. LSSO-r32 uses 0.454G MACs and takes 1.95 ms. Peak memory is closer to optimized MHA than the dense $O(N^2)$ formula would suggest because the benchmark uses an optimized SDPA path related to memory-efficient exact attention kernels [7]. The practical advantage is therefore clearest in arithmetic cost and long-sequence latency, rather than peak memory at every sequence length.

5.7 Vision and Diffusion Boundary Results

Table 5 reports ViT-style [10] vision encoder results on CIFAR-100 [12] and an ImageNet subset [8]. The controlled vision runs use RandAugment [6], Mixup [28], and CutMix [26], following data-efficient vision-transformer training practice [20]. CIFAR-100 uses 32×32 images, patch size 2, $D = 96$, depth 3, 6 heads, batch size 256, and 100 epochs. Its augmentation consists of random crop and horizontal flip, RandAugment, Mixup 0.2, CutMix 0.5, and label smoothing 0.1. ImageNet-100 uses 224×224 images, patch size 8, $D = 256$, depth 8, 8 heads, batch size 256, and 120 epochs, with Mixup 0.8, CutMix 1.0, RandAugment, and label smoothing 0.1. Both use AdamW, cosine decay, and identical settings across mixers within a dataset.

On CIFAR-100, the LSSO models trade accuracy for cost and do not match Nyströmformer. On ImageNet-100, MHA is about 1.7 points stronger in the single-seed run, whereas LSSO cuts mixer MACs by roughly three to four times. Vision therefore gives a more mixed picture than retrieval: the operator remains competitive, but its efficiency gain is not free.

Table 5: Vision encoder results. MACs cover all mixer layers in the encoder. CIFAR-100 is 3 seeds; ImageNet-100 is one seed.

Dataset	Model	MACs (G)	Top-1	Top-5
CIFAR-100	MHA	0.0665	55.40 ± 0.27	–
CIFAR-100	Nyströmformer	0.0389	59.98 ± 0.58	–
CIFAR-100	LSSO-r16	0.0250	54.79 ± 0.14	–
CIFAR-100	LSSO-r32	0.0388	55.38 ± 0.18	–
ImageNet-100	MHA	4.162	82.26	95.54
ImageNet-100	LSSO-r16	1.093	80.54	94.88
ImageNet-100	LSSO-r32	1.391	80.58	94.76

We also tested a pure-LSSO latent diffusion backbone on ImageNet-100. The available experiment measures only noise-prediction validation MSE, not FID, KID, or sample quality. LSSO reaches a similar optimization regime at lower theoretical mixer cost, but this evidence is not sufficient to claim that it can replace attention in a generative model.

6 Discussion

The experiments suggest that LSSO is best viewed as a bidirectional global correction operator, not as a cheaper approximation to softmax attention. This distinction appears especially useful in retrieval, where an encoder must compress evidence from the whole sequence into a representation but does not need causal next-token routing. A small learned subspace can influence all tokens through the solve without constructing an all-pairs value-routing matrix. This interpretation is supported by the retrieval results in Table 1, the zero-shot transfer results in Table 2, and the nonzero correction ratios in Figure 1.

The same construction is less natural for causal decoding. Applying a causal mask breaks the simple low-rank Woodbury form, and an autoregressive extension would require prefix-restricted solves or a different cached state. The diffusion experiment points to a second boundary: global semantic mixing alone may not recover the fine-grained routing that generation demands.

7 Limitations

Rank is a genuine capacity choice: some tasks occupy only a compact subspace, whereas SciFact uses much more of rank 32 (Figure 1 and Table 4). The vision and diffusion results are also narrower than the retrieval evidence. ImageNet-100 has only one seed, and the diffusion study lacks sample-quality metrics. These experiments establish useful operating points, not universal superiority over attention or other efficient mixers.

8 Conclusion

LSSO replaces pairwise value routing with a learned low-rank global solve. The positive shift makes the operator well posed, and Woodbury makes the solve practical when $r \ll N$ (Eq. (2) and Appendix A). In our experiments this inductive bias is particularly effective for retrieval: LSSO preserves MHA-level transfer while substantially reducing mixer computation (Tables 1 and 2). Vision and diffusion results are less decisive, which helps delimit the present claim: LSSO is a promising efficient mixer for bidirectional encoders, rather than a general replacement for attention.

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A Full Woodbury Derivation

For $\gamma > 0$, using the Woodbury identity with base matrix $A_0 = \mu I_N$, low-rank factors $P = U$ and $Q = U^\top$, and middle matrix $M = \gamma I_r$ gives

$$(\mu I + \gamma U U^\top)^{-1} = \mu^{-1} I - \mu^{-1} U (\gamma^{-1} I + \mu^{-1} U^\top U)^{-1} U^\top \mu^{-1}.$$

Multiplying the small inverse by γ avoids an explicit γ^{-1} and gives the implementation form

$$Y = \mu^{-1} C - \mu^{-1} U (I + \gamma \mu^{-1} U^\top U)^{-1} (\gamma \mu^{-1} U^\top C),$$

which remains valid at $\gamma = 0$.